Timothy (Tim) Prater, MS

Principal Engineer | Systems Architect | Networking, and Engineering Leader

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EXECUTIVE PROFILE

Principal technology leader with 19 years of experience spanning carrier-grade networking, mobile core, systems integration, solution architecture, pre-sales leadership, cloud architecture, and customer-facing technical strategy. Career foundation built through hands-on engineering across LTE/EPC, packet core routing, HA production operations, and packet-level troubleshooting, then expanded into architecture leadership, private wireless strategy, RFx ownership, delivery governance, and executive engagement.

Known for working comfortably at the intersection of technical depth and business execution: defining architectures, building delivery-ready low-level designs, shaping statements of work, leading customer discovery, supporting escalations, and creating repeatable service offerings. Background includes private LTE and private 5G, MVNO enablement, IoT and M2M, cloud-connected architectures, enterprise networking, security design, and cross-functional leadership across Sales, Engineering, Product, Legal, Operations, and customer teams.

This document is intentionally detailed and role-encompassing. It is meant to function as a master professional CV that captures technologies, platforms, methods, and representative scope by role throughout my entire 19-year tenure.

CORE CAPABILITY AREAS

Architecture & Technical Leadership: cross-functional leadership, HLD/LLD, technical governance, incident leadership, mentoring, executive communication, production readiness, failover planning, high availability
Private Wireless & Mobility: private LTE, private 5G, LTE/EPC, 5GC, MVNO enablement, APN design, roaming, IoT/M2M, carrier interconnect, mobile core integration, acceptance testing
Networking & Security: Cisco routing and switching, IPv4, OSPF, BGP, MPLS, VRF, VLAN, GRE, IPsec, ACLs, NAT, QoS, DNS, DHCP, VPNs, network segmentation, defense in depth, access controls
Cloud, Automation & Data: AWS, Lambda, API Gateway, DynamoDB, S3, SNS, SES, EventBridge, IAM, CloudWatch, Amplify, SQL, PL/SQL, Python, R, Linux, Unix, ETL, workflow automation

SELECTED STRATEGIC INITIATIVES AND REPRESENTATIVE TECHNICAL WORK

MVNO / IoT Aggregator Enablement - 3GPP / GSMA-Compliant LTE Architecture | 2024 - 2025

- Led end-to-end architecture and delivery planning for a standards-aligned LTE integration model that allowed MVNOs, private cellular customers, universities, and related partners to interconnect with carrier network capabilities and leverage roaming and peering agreements.
- Created a repeatable blueprint supporting both AWS-hosted and on-premises LTE core models and vendor combinations including Nokia, Ericsson, and third-party environments.
- Drove work from technical strategy and architecture definition through executive alignment, RFx support, and Acceptance Test Plan creation, contributing to 30+ signed contracts with terms of 1 to 5 years and deal values roughly between $500K and $3M.

Technical focus: 3GPP, GSMA, LTE architecture, IPX, EPC MVNO enablement, roaming, APN design, AWS-hosted LTE cores, Nokia, Ericsson, acceptance testing, architecture governance.

Professional Services Portfolio Development | 2025

- Designed a portfolio of professional services offerings including consulting, audits, architecture assessments, and security assessments so Sales could package repeatable high-value engagements instead of relying only on ad hoc engineering support.
- Created reusable scope language, delivery framing, technical positioning, and customer-facing artifacts that supported multiple engagements valued at approximately $250K-$700K over 2-6 month durations.

Technical focus: Service portfolio design, scope development, customer-facing collateral, architecture assessments, security assessments, delivery models, and reusable sales-engineering artifacts.

Nationwide LTE Ecosystem Audit | 2025

- Led an end-to-end LTE ecosystem audit to identify configuration drift, topology inconsistencies, failover weaknesses, and performance outliers across a broad production environment.
- Established golden-parameter standards and improved topology documentation; remediation efforts improved failover behavior and reduced mean time to restore by approximately 40% during network events.

Technical focus: LTE audits, topology review, standards baselining, failover analysis, operational documentation, remediation planning, and restoration improvement.

Multi-Tenant Notification & Review Platform | 2025

- Architected a multi-tenant customer notification and review platform that ingested engineering work requests, enabled internal validation workflows, and generated standardized customer communications, improving consistency, reducing manual effort, and ensuring accurate, scalable message delivery across enterprise accounts.

Technical focus: Microsoft Power Platform, including Power Apps, Power Automate, SharePoint Online and SharePoint Lists, Outlook/Teams notification integration, approval workflows, communication templates, and internal validation workflow automation.

Multi-Tenant SaaS Tracking Platform | 2025

- Designed and led development of a multi-tenant, SaaS-style tracking platform to manage white-glove customer engagements, providing centralized visibility into onboarding status, delivery milestones, and customer-specific activities to improve coordination across internal teams and enhance customer experience.

Technical focus: AWS multi-tenant application architecture using Amazon API Gateway, AWS Lambda, Amazon DynamoDB, Amazon S3, Amazon Cognito, IAM, CloudWatch, and event-driven workflow design to support onboarding tracking, milestone visibility, account-level activity management, and centralized operational reporting.

PROFESSIONAL EXPERIENCE

T-Mobile - Principal Engineer, System Architecture | Aug 2025 - Feb 2026 | Knoxville, TN - Remote

Fixed-term principal role following the U.S. Cellular acquisition, focused on transition continuity, migration planning, and technical coordination as legacy environments and customer obligations moved into T-Mobile operating models.

• Led cross-functional transition efforts spanning engineering, operations, legal, program stakeholders, and customer-facing teams to align migration assumptions, service dependencies, communications, and execution sequencing during a complex post-acquisition integration.

• Directed technical alignment across legacy U.S. Cellular LTE / EPC, private cellular, and related support environments, mapping services into T-Mobile target-state processes and infrastructure while balancing migration

risk, customer commitments, operational readiness, and architectural constraints.
• Provided leadership on architecture and service-delivery decisions involving AWS-connected environments, access controls, VPC design, network segmentation, and L2/L3 Cisco connectivity, helping multiple teams converge on workable transition paths for both customer solutions and internal support workflows.
• Served as the senior technical bridge between legacy knowledge and target-state execution, leading working sessions, clarifying service behaviors and architectural dependencies, documenting transition assumptions, and helping teams navigate new tools, support paths, and governance expectations.
• Guided matrixed teams through ambiguity in a high-visibility integration period by translating operational realities into actionable engineering decisions and maintaining alignment across business, technical, and customer-impact considerations.

Representative technologies and methods: AWS, VPC design, security and access controls, L2/L3 Cisco networking, migration planning, systems integration, service delivery coordination, operational readiness, stakeholder management, and post-acquisition transition support.

U.S. Cellular – Principal / Lead Enterprise Solutions Engineer | May 2021 - Aug 2025 | Knoxville, TN - Remote

Principal-level leadership role spanning architecture, technical strategy, customer engagement, RFx ownership, and team leadership for private wireless, enterprise networking, IoT, and AWS-connected opportunities.

- Led cross-functional teams of senior solutions engineers, senior integration engineers, project managers, and customer relationship managers in support of complex private wireless, enterprise networking, and cloud-connected solutions for enterprise, utility, government, and MVNO-related opportunities.
- Served as principal technical author for RFP, RFI, and RFQ responses, translating customer, security, regulatory, and operational requirements into delivery-ready solution narratives, diagrams, assumptions, implementation models, and technical compliance mappings that influenced more than $10M annually in business.
- Acted as the primary SME for executive and customer-facing presentations, leading discovery sessions, architecture reviews, and strategy discussions covering private LTE, private 5G, IoT / M2M connectivity, roaming models, APN design, high-availability approaches, network segmentation, and cloud integration tradeoffs.
- Developed architecture and deployment models for private wireless and enterprise environments including GRE and IPsec tunnel designs, segmented customer connectivity, failover strategies, smaller-footprint customer builds, and interconnection patterns linking carrier, enterprise, and on-premises environments.
- Built internal SaaS-style tools and workflow automations to improve provisioning visibility, communication cadence, and high-priority customer tracking; used AWS services to create lightweight operational tooling that reduced manual coordination overhead and improved transparency for engineering and sales teams.
- Created runbooks, onboarding plans, training content, and individualized development plans used across the organization; formalized delivery artifacts such as low-level designs, implementation guidance, and acceptance-test materials so solution work could scale more predictably.
- Guided post-sales technical governance through infrastructure audits, architecture assessments, security assessments, implementation planning, and Acceptance Test Plan development to reduce downstream delivery risk and improve handoff quality.

Representative technologies and methods: AWS, Lambda, API Gateway, DynamoDB, S3, SNS, SES, EventBridge, IAM, CloudWatch, private LTE, private 5G, LTE, 5G, EPC, 5GC, APN design, roaming strategy, Cisco GRE, IPsec, secure network segmentation, HLDs, LLDs, RFx technical authorship, runbook development, training development, acceptance testing, and cross-functional delivery leadership.

U.S. Cellular - Senior Business Solutions Engineer | Oct 2017 - May 2021 | Knoxville, TN - Remote

Customer-facing solution engineering role focused on IoT, M2M, and connected business solutions across design, feasibility analysis, integration, implementation support, and production readiness.

- Served as technical SME for IoT and M2M opportunities by translating customer requirements into architectures, low-level designs, implementation plans, and supportable integration patterns across network, application, and provisioning boundaries.
- Led discovery, requirements analysis, and detailed design development for solutions involving API and interface integration, onboarding and activation flows, provisioning workflows, secure connectivity models, data movement, and operational support considerations.
- Worked through feasibility reviews, testing strategy, and implementation planning for new or modified services; supported product and solution certification efforts by creating test plans, validating integration behavior, isolating defects, and coordinating with engineering and vendor teams through resolution.
- Applied architecture and security discipline to solution design by incorporating least privilege, segmentation, role-based access, and NIST-minded controls into guidance for customer-facing and internally delivered solutions.
- Produced customer-facing technical documentation and architecture walk-throughs that clarified constraints, dependencies, support expectations, and implementation risk for both technical and non-technical stakeholders.

Representative technologies and methods: IoT, M2M, API and interface integration, onboarding and activation workflows, provisioning design, low-level architecture documents, secure network design, testing and certification planning, network and application troubleshooting, NIST, Defense in Depth, Principles of Least Privilege security practices, and customer-facing architecture presentations.

U.S. Cellular - Mobile Core Engineer | Jun 2014 - Oct 2017 | Knoxville, TN – Hybrid

Carrier-grade operations and engineering role supporting packet core and LTE environments with heavy emphasis on production stability, fault isolation, protocol-level troubleshooting, and nationwide service integrity.

- Maintained and supported carrier-grade mobile core infrastructure across LTE and legacy environments with focus on production stability, service integrity, network health, performance analysis, and rapid isolation of service-impacting issues.
- Worked extensively in Cisco routing environments troubleshooting end-to-end service behavior across IPv4, TCP/IP, UDP, OSPF, BGP, DNS, DHCP, NAT, ACLs, QoS, VPNs, and IPsec in support of packet core services and supporting transport paths.
- Performed deep packet and protocol analysis using Wireshark and PCAPs to trace signaling and transport flows, validate message paths, and identify root cause across Diameter, GTP, DNS, DHCP, session establishment, and related mobility service failures.
- Led Tier 4 and 24x7 incident response for critical events involving core platforms, routing paths, failover scenarios, load-balancing behaviors, and vendor-integrated systems; coordinated restoration activity and technical updates during major incidents.
- Contributed to optimization of nationwide infrastructure and supported environments that achieved approximately 99.99% service availability across data and signaling systems.

Representative technologies and methods: LTE, EPC, SGW, PGW, MME, HSS, Linux, Unix, Cisco routing, IPv4, TCP/IP, UDP, OSPF, BGP, DNS, DHCP, NAT, ACLs, QoS, VPN, IPsec, Diameter, GTP-U/C, Wireshark, PCAP analysis, Netcool, Netscout, Spirent, F5 load-balancing, and Tier 4 incident leadership.

U.S. Cellular - Associate Enhanced Data Services Engineer | Feb 2012 - Jun 2014 | Knoxville, TN - Hybrid

Data engineering and application support role focused on ETL, enterprise data movement, operational reporting, and production support across warehouse and business-system environments.

- Developed and supported data integration and data lake solutions moving data between business systems, enterprise data warehouses, data marts, and operational data stores; handled extraction, transformation, cleansing, and load workflows supporting reporting and downstream applications.
- Worked with project managers, business analysts, and technical architects to translate requirements into practical ETL designs, data migration plans, data-cleansing logic, and data-modeling solutions.

- Built and maintained code from detailed design specifications, supported unit, system, and integration testing, and provided production support for application defects, query issues, and data-quality problems.

Representative technologies and methods: Oracle, SQL, PL/SQL, ETL, data migration, data cleansing, data modeling, Informatica, Talend, Hadoop, MongoDB, Git, SOA concepts, SDLC, Agile delivery, and production application support.

U.S. Cellular - Roaming and Data Technical Specialist | May 2006 - Feb 2012 | Knoxville, TN

Progressive frontline-to-advanced support role for wireless data, roaming, messaging, and provisioning systems across internal platforms and carrier interconnects.

- Provided advanced customer and operations support for wireless data and roaming issues, troubleshooting across billing systems, switches, voicemail, messaging services, mobile data functionality, provisioning systems, and authentication platforms.
- Worked across SMSC, MMSC, HLR, VLR, SIM provisioning, AAA data servers, and related environments to resolve customer-impacting issues and coordinate with internal teams, vendors, and domestic and international carriers.
- Supported roaming configuration changes, audits, and translation-related updates, acting as a liaison between customers, operational teams, network teams, and vendor personnel while maintaining accurate documentation and case tracking.

Representative technologies and methods: SMSC, MMSC, HLR, VLR, SIM provisioning, AAA, roaming support, carrier interconnect coordination, Remedy / BMC workflows, HTTP, POP, SMTP, mobile data troubleshooting, switch, customer documentation, and escalation handling.

TOOLS, PLATFORMS, AND PROTOCOLS BY DOMAIN

Networking and Transport: Cisco IOS and enterprise / carrier routing concepts, IPv4, TCP/IP, UDP, OSPF, BGP, MPLS, Segment Routing, VRF, VLAN, GRE, NAT, ACLs, QoS, VPN, and IPsec.

Private Wireless and Mobility: Private LTE, private 5G, LTE/EPC, 5GC, APN design, roaming, MVNO enablement, IoT/M2M architectures, Diameter, GTP-U/C, DNS, DHCP, and segmented traffic models.

Cloud and Automation: AWS, Lambda, API Gateway, DynamoDB, S3, SNS, SES, EventBridge, IAM, CloudWatch, Cognito, Amplify, workflow automation, and internal SaaS-style operational tooling.

Programming and Development: Python, Git, object-oriented programming, functional programming, AI agent integration, retrieval-augmented generation (RAG), scripting, automation logic, and debugging.

Operational Tooling: Linux, Unix, Cisco CLI, Wireshark, PCAP analysis, Netcool, Netscout, Spirent, Remedy/BMC, NetAct, F5, Palo Alto, SSH, SFTP, SNMP, Telnet, and LDAP.

Vendor Exposure: Nokia, Cisco, Affirmed, F5, Ericsson, Alcatel-Lucent / Nokia, Oracle, Mavenir, Innopath, Thales, and Cradlepoint in customer or solution contexts.

EDUCATION, CERTIFICATIONS, AND RECOGNITION

- Master of Science, Information Technology Network Management - Colorado Technical University
- Bachelor of Science, Computer and Information Systems Security; Associate of Applied Science, Computer Networking Systems - International Telephone & Telegraph Technical Institute
- Alcatel-Lucent Network Routing Specialist I (formerly held)
- Cisco Certified Network Professional (formerly held)
- Leadership Academy - U.S. Cellular (top of class)
- Common Purpose Award (2x) for cross-organizational collaboration and removal of technical roadblocks affecting business outcomes
- Consistently earned Exceeds Expectations or Far Exceeds Expectations performance ratings